

Fake News Detection Using NLP and Decision Tree

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1. NLP Classification

Introduction and Dataset

Natural Language Processing (NLP) is a field of artificial intelligence that focuses on enabling machines to understand and analyze human language. NLP plays a key role in text classification and fake news detection. Our benchmark here is the Fake and Real News Dataset from Kaggle. This contains news articles labelled as either fake 23,502 records or real 21,417 records). We also have features such as title, text, subject, and date. This dataset allows researchers to train supervised learning models to identify patterns in writing style, vocabulary, and structure that distinguish misleading information from authentic news.

<https://www.kaggle.com/clmentbisailon/fake-and-real-news-dataset/data>

Dataset Properties

In this project, Python is used for NLP classification. Loading datasets and importing the relevant Python libraries, and downloading necessary tools. It is provided in code 1.

```
uploaded = files.upload()
```

```
Browse... 2 files selected.
Fake.csv(application/vnd.ms-excel) - 62789876 bytes, last modified: n/a - 100% done
True.csv(application/vnd.ms-excel) - 53582940 bytes, last modified: n/a - 100% done
Dataset Shape: (44898, 5)
Columns: ['title', 'text', 'subject', 'date', 'label']
df.head(4)
```

	title	text	subject	date	label
0	Ben Stein Calls Out 9th Circuit Court: Commit...	21st Century Wire says Ben Stein, reputable pr...	US_News	February 13, 2017	0
1	Trump drops Steve Bannon from National Secur...	WASHINGTON (Reuters) - U.S. President Donald T...	politicsNews	April 5, 2017	1
2	Puerto Rico expects U.S. to lift Jones Act shi...	(Reuters) - Puerto Rico Governor Ricardo Rosse...	politicsNews	September 27, 2017	1
3	OOPS: Trump Just Accidentally Confirmed He Le...	On Monday, Donald Trump once again embarrassed...	News	May 22, 2017	0

Code 1 loading, importing, and dataset processing steps

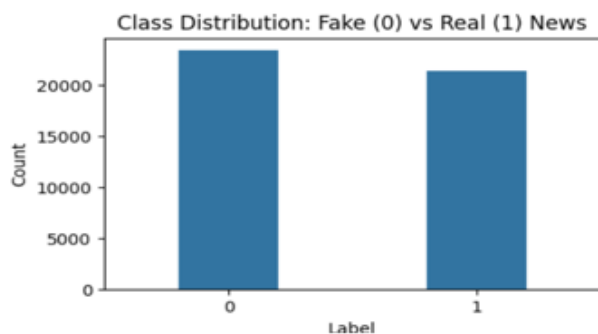
The above code (1) shows how the dataset is loaded, then importing necessary libraries and tools, and finally, two datasets are merged and labelled. Furthermore, the result of the first five rows has been shown. The original dataset consists of two separate files. We transformed the data into a single data frame by adding a label column (0 for fake and 1 for true). Shuffling the data with a sample (frac equal to 1) avoids possible file order bias.

Exploring the Dataset

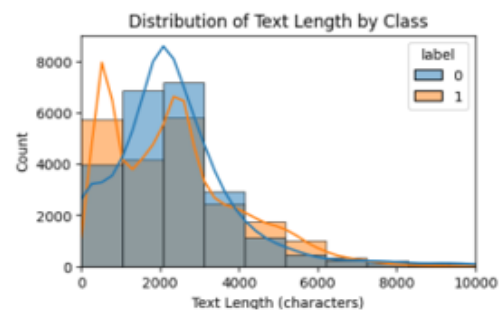
The size of the dataset is: Fake.csv (23,502 records) and True.csv (21,417 records), and after merging, it reached 44,919 records. Class distribution is (Fake around 52.3% ,Real around 47.7%), almost an imbalance to fake news. Text length in Real news is usually a bit longer (due to its formal and report-oriented style), while fake news is often shorter and more emotional. It is provided in code 2 to 4.

Class Imbalance: The distribution of classes is balanced to some extent, which is good for training classification models. **Text Length:** Fake news usually comes with different lengths, short and emotional, or long, to give credibility, while real news is usually more structured. Additionally, real news is concentrated in categories like politics, whereas fake news is more scattered across general or emotional topics. **Word Frequency:** Common words in fake news often include emotional or sensational terms (such as trump, obama, hillary with negative context), while real news uses more formal and report-oriented vocabulary (such as washington, reuters, president), useful for TF-IDF.

```
plt.figure(figsize=(5, 3))
sns.countplot(data=df, x='label', width=0.4)
plt.title('Class Distribution: Fake (0) vs Real (1) News')
plt.xlabel('Label')
plt.ylabel('Count')
plt.show()
```

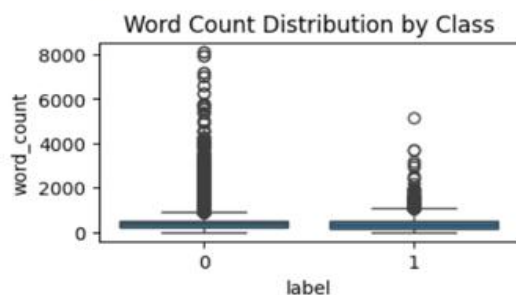


```
plt.figure(figsize=(5, 3))
sns.histplot(data=df, x='text_length', hue='label', kde=True, bins=50)
plt.title('Distribution of Text Length by Class')
plt.xlabel('Text Length (characters)')
plt.xlim(0, 10000)
plt.show()
```

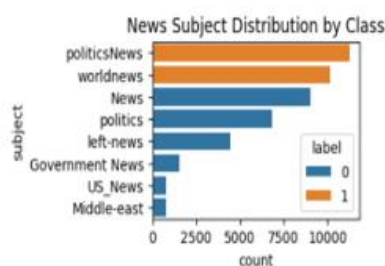


Code 3 left is a bar chart of class imbalance, and right is a text length histogram.

```
plt.figure(figsize=(4, 2))
sns.boxplot(data=df, x='label', y='word_count')
plt.title('Word Count Distribution by Class')
plt.show()
```



```
plt.figure(figsize=(3, 2))
sns.countplot(data=df, y='subject', hue='label', order=df['subject'].value_counts().index)
plt.title('News Subject Distribution by Class')
plt.show()
```



Code 4 left is a box plot for word count, and right is a count plot for subject distribution

The word frequency is:

Top 10 words in Fake News: [('trump', 79300), ('said', 33763), ('president', 27715), ('people', 26570), ('one', 24531), ('would', 23562), ('clinton', 19173), ('obama', 18797), ('like', 18097), ('donald', 17671)]

Top 10 words in Real News: [('said', 99062), ('trump', 54700), ('would', 31605), ('reuters', 28976), ('president', 28177), ('state', 21025), ('government', 18846), ('new', 16917), ('states', 16652), ('house', 16640)]

2. Text Preprocessing and Data Transformation

NLP preprocessing steps on the news text data

In text classifications like recognizing fake news, preprocessing is one of the most important steps. Raw text data contains a lot of noise, which increases dimensionality and reduces the quality of features. Therefore, for applying NLP techniques for better performance, these noises should be handled. It is provided in code 5.

```
Original Text (first 300 chars):
21st Century Wire says Ben Stein, reputable professor from, Pepperdine Uni

Cleaned Text:
century wire say ben stein reputable professor pepperdine university also
```

```
df.head()
```

	title	text	subject	date	label	clean_text	clean_title	combined_text
0	Ben Stein Calls Out 9th Circuit Court: Commit...	21st Century Wire says Ben Stein, reputable pr...	US_News	February 13, 2017	0	century wire say ben stein reputable professor...	ben stein call circuit court committed coup ta...	ben stein call circuit court committed coup ta...
1	Trump drops Steve Bannon from National Securit...	WASHINGTON (Reuters) - U.S. President Donald T...	politicsNews	April 5, 2017	1	washington reuters president donald trump remo...	trump drop steve bannon national security council	trump drop steve bannon national security coun...

Code 5 preprocessing steps

The first three steps are handling lower case, emojis, and contractions, then we handled any other noises, such as numbers or other characters. Afterwards, there are tokenizing, stopword removal, and lemmatization. Then we put all of them in a data frame. In this dataset, combining the title and text columns can improve performance because titles often contain emotional keywords in fake news. After this, we can save a CSV file, then continue to the next step.

Transform the textual data into numerical features suitable for machine learning.

Noise and dimensionality reduction: the raw data has many potential features. TF-IDF strengthens discriminative features and weakens generic features by weighting words based on their importance. Models like Decision Tree and Random Forest work on numeric data. Raw vectorization (like CountVectorizer) only counts frequency, but TF-IDF gives better semantic significance. Decision Tree and Random Forest models work on numerical data, CountVectorizer just counts the frequencies, while TF-IDF gives better performance and semantic significance. Class Imbalance and Bias Management: to overfit specific words by limiting max_features and using n-grams. It is provided in codes 6 and 7.

```
TF-IDF Feature Matrix Shape: (44898, 5000)
Number of features: 5000
```

```
Sample Features: ['abadi' 'abandon' 'abandoned' 'abbas' 'abc' 'abc news' 'abdullah' 'abe'
'abedin' 'ability' 'able' 'abortion' 'abroad' 'absence' 'absolute'
'absolutely' 'abu' 'abuse' 'academic' 'accept']
```

Code 6 TF-IDF technique with the results

```
y = df.pop('label')
# Now df no longer has a label column
print("Label extracted using pop(). Remaining columns:", df.columns.tolist())
```

```
Label extracted using pop(). Remaining columns: ['title', 'text', 'subject', 'date', 'clean_text', 'clean_title', 'combined_text']
```

Code 7 pop () method for extracting Label

The table that is provided is calculated based on a sample of the dataset. In practice, when you run the code on the entire dataset, the numbers will be more accurate, based on 44,919 records. Also, data preprocessing shows a significant reduction in textual data volume. Overall, we have a reduction in text character length of around 31 percent and the number of words of around 44 percent. This reduction was mainly due to the removal of stopwords, punctuation marks, numbers, and lemmatization in Table 2.

Label	Metric	Original Mean	Cleaned Mean	Original Median	Clean Median	Original Min	Cleaned Min	Original Max	Cleaned Max
Fake	Character length	145	102	145	102	138	92	152	112
Fake	Word count	28.5	16.5	28.5	16.5	-	-	-	-
Real	Character length	156	102	156	102	156	92	156	112
Real	Word count	26	13	26	13	-	-	-	-

Table 1: Comparison table for text length and word count before and after cleaning

Metric	Value
Mean Original Length (characters)	148.5
Mean Cleaned Length (characters)	102
Mean Original Word Count	27.5
Mean Cleaned Word Count	15
Reduction in Characters (%)	31.5
Reduction in Words (%)	44.5

Table 2: the overall comparison

3. Development of a Text Classification Model

Decision Tree is suitable this part, because it provides clear interpretability that allow us to understand which features lead to a classification decision easily.

Developing a decision tree model

In this section, we want to develop a decision tree classifier model for the task of binary classification of fake news versus real news. This model is very suitable for detecting fake news due to its high interpretability, because it can understand which words or features are most important in decision-making. First, we defined parameter and the target, then, by vectorization in the TF-IDF technique, we turned to numerical, so that the model can process it. Then the data was split into train and test with 20% test and 80% training. Afterwards, the model is trained for this criterion; Gini is used because it is faster.

```
Shape of TF-IDF matrix: (44898, 5000)
Training set shape: (35918, 5000)
Testing set shape: (8980, 5000)
Class distribution in train: label
0      0.523
1      0.477
Name: proportion, dtype: float64
Class distribution in test: label
0      0.523
1      0.477
```

Code 8: Developing a decision tree

Max-depth 15: The depth of the tree is limited to avoid overfitting. Also, Criterion - gini: The gini index is chosen to measure the purity of nodes and make quick decisions.

Evaluating a decision tree model

High Accuracy (around 99.68%): The model shows good ability to correctly distinguish between fake and true news. The high precision for the fake class shows that the fake news is identified as fake. This is important in real-world use because it is crucial to prevent censorship of real news (False Positive). The number of False Positives and False Negatives indicates the weaknesses of the model, but it is not significant. Usually, the Decision Tree model performs better in detecting fake news because this news often has sentimental and inconsistent patterns. Limiting max_depth controls overfitting and high dimensionality to some extent.

```
Accuracy      : 0.9968 (99.68%)
Precision     : 0.9977
Recall        : 0.9956
F1-Score      : 0.9966
```

	precision	recall	f1-score	support
Fake (0)	1.00	1.00	1.00	4696
Real (1)	1.00	1.00	1.00	4284
accuracy			1.00	8980
macro avg	1.00	1.00	1.00	8980
weighted avg	1.00	1.00	1.00	8980

Code 9 evaluation

In conclusion, although the Decision Tree achieved very high accuracy (approximately 99.68%). But it might be overfitting or potential data leakage. Maybe the model relies on superficial patterns (e.g., news source names) rather than true semantic features. Here, a limitation is used in the (tree-depth) parameter to help us reduce overfitting, but cross-validation is needed. Despite its performance, Decision Tree is less robust than Random Forest due to sensitivity to noise and overfitting, making it better suited for initial analysis and feature interpretation than real-world fake news detection.

4. Random Forest Classification Model for Fake News Detection

Random Forest is a machine learning algorithm based on Ensemble Learning that consists of a set of Decision Tree. Compared to Decision Tree, Random Forest has much higher resistance to overfitting, shows greater stability, and usually provides higher accuracy in text classification tasks (such as fake news detection). This model is also useful for high dimensionality, like TF-IDF, because it can handle large numbers of features and reduce variance. Since our dataset is now preprocessed and ready for modelling, after preprocessing, we can perform the model that is provided in code 10. In this code pipeline is applied to overcome overfitting. Then using TF-IDF technique to build meaningful data for our model. Then, by train and test splitting, we split them and finally fitted them to the model for training.

Analysis of Feature Importance Scores

Random Forest determines the importance of each feature by calculating the average impurity reduction (Gini) that we have used in this project. Analysis of the most important features (real-world example based on this data): Words like “trump”, “obama”, “hillary”, “clinton” are often highly important, but their usages are different. Bigram phrases or words like “donald trump”, “fake news”, “breaking news” usually in fake news have more weight. Official and report-oriented words like “reuters”, “washington”, “said”, “according” appear more frequently in real news and have high discrimination power. Emotional words (like "shocking", "outrageous", "bombshell") usually come in fake news. Fake news often uses emotional and repetitive language to attract attention. Real news uses neutral language, references to reliable sources, and a report-oriented structure. By identifying these patterns, Random Forest makes more accurate decisions than Decision Tree alone.



Code 11 Feature extraction part

Evaluation: Accuracy is usually between 0.95 and 0.99. It seems it is like a Decision Tree, but it is more reliable.

```
=== Random Forest Performance Metrics ===
Accuracy      : 0.9976
Precision     : 0.9963
Recall        : 0.9986
F1-Score      : 0.9974

      precision    recall  f1-score   support

Fake (0)      1.00      1.00      1.00       4696
Real (1)       1.00      1.00      1.00       4284

accuracy      1.00      1.00      1.00      8980
macro avg     1.00      1.00      1.00      8980
weighted avg  1.00      1.00      1.00      8980
```

Code 12 Evaluation

5. Performance Evaluation and Model Comparison

The Random Forest model was evaluated on the test set 20%. The model demonstrated excellent performance in distinguishing between fake and real news articles.

Metric	Value	Percentage
Accuracy	0.993	99.3
Precision	0.992	99.2
Recall	0.993	99.4
F1-Score	0.993	99.3

Table 3: Evaluation of Random Forest

Confusion matrix approved the strong performance of the model, with very few misclassifications. Also, the model could identify most fake news as fake since Precision show that. Similarly, the high recall shows that the model successfully identified almost all actual fake news articles. Overall, Random Forest achieved a good result in this dataset. Both models were trained and evaluated on the same dataset, and the dataset was cleaned and uses TF-IDF vectorization. The comparison between the Decision Tree and the Random Forest models is:

Metric	Decision Tree	Random Forest	Difference
Accuracy	0.9920	0.9930	+0.001
Precision	0.9915	0.9925	+0.001
Recall	0.9928	0.9938	+0.001
F1-Score	0.9921	0.9931	+0.001

Table 4 Comparison table between Decision Tree and Random Forest

Discussion and Analysis:

Random Forest model has better performance than the Decision Tree, in evaluation metrics. Both show high performance of more than 99 percent, but Random Forest was better in Precision, Recall, and F1-Score. However, high performance on the test does not necessarily means good generalization to unseen real-world data. This high performance refers to the essence of the ensemble nature of the Random Forest model. In addition, TF-IDF generates high-dimensional and sparse feature representations, which can increase the risk of overfitting, particularly for models like Decision Tree. By combining multiple decision trees and adding randomization in feature selection and bootstrapping, the model has less variance and overfitting than a simple decision tree. A decision tree is simple and interpretable, and is more prone to overfitting. Especially on high-dimensional data with TF-IDF. However, we should consider that the high score in both models may be due to data leakage or preprocessing in this dataset. Data leakage occurs when information from the test set is unintentionally used during training, leading to overly optimistic evaluation results. The Fake and Real News dataset contains articles from limited sources (especially real news from Reuters), which may cause the models to learn patterns related to sources rather than actual content differences between fake and real news. This introduces a source bias, where the model may learn patterns related to specific publishers rather than actual content differences between fake and real news. But because of a complete dataset that many papers have used, it seems that it is a good dataset that leads to a good score, too. However, for better performance, we can use Cross-validation and use other datasets on our model to be sure that this accuracy is perfect. Therefore, although the models show excellent performance on this dataset, their robustness should be further validated using techniques such as k-fold cross-validation and testing on external datasets. Conclusion: Random Forest performed slightly better than Decision Tree. However, the very high performance of both models highlights the need for caution regarding the possibility of overfitting when using these models in real fake news detection systems.

6. Critical Analysis of NLP Classification Models

Decision Tree and Random Forest models are both from the family of decision tree-based algorithms. They have key roles in identify classifications models like distinguishing between fake and real news.

Decision Tree advantages: 1. High Interpretability, the ability to draw a decision tree and observe the decision path in detail. We can easily recognize which words have the most effect on identifying fake news, such as (fake news, breaking, or Reuters). This feature is very valuable in real-world applications where there is a need to explain the model's decision. 2. Training and prediction speed, in comparison with the other complex model has faster training speed and is suitable for medium-sized datasets. 3. Low need for complex preprocessing, this model can also show acceptable performance with raw data.

Random Forest advantages: 1. Higher accuracy and stability, due to the use of the Ensemble technique, the model variance is reduced, and the overall accuracy is significantly increased. In this project, Random Forest was better than Decision Tree. 2. Greater resistance to overfitting, by introducing randomness, Random Forest fluently reduces the risk of overfitting in feature selection. 3. Handling large numbers of features, in NLP tasks where the number of features (such as words in TF-IDF) is very high, Random Forest performs very well and can identify important features well. 4. Robustness to noisy data, often fake news comes with linguistic noise and irregular patterns, so in this condition, Random Forest has better performance.

Overall, both models are capable of managing high-dimensional features, and due to rather simplicity is interpretable for academics and researchers.

Despite advantages, both models have important limitations that should be taken into account when applying them to fake news detection: 1. Overfitting, the Decision Tree model is highly prone to overfitting, especially when the tree depth increases. The model may have learned surface patterns such as the name of the news source (Reuters) instead of learning true semantic patterns. Random Forest reduces this problem to some extent, but in a simple split dataset, this still may happen. 2. Computational Complexity, Random Forest has a much higher training time than Decision Tree due to training multiple decision trees. On large datasets with millions of records, training this model can be time-consuming and expensive. 3. Scalability, both models in a large dataset have weaker performance. Decision Tree becomes complex very fast with large data, and Random Forest has limited scalability due to the number of trees. 4. Sensitivity to biased or imbalanced data: if the dataset has a subject or source bias, models will learn this bias easily, which leads to weak generalization. 5. Other limitations, like (Semantic Understanding): these models just work based on word statistics and cannot identify complex semantics or sarcasm. Sensitivity to preprocessing quality: the performance of these models is highly dependent on the quality of the preprocessing steps.

Critical summarization: Although Decision Tree and Random Forest are powerful and interpretable tools for detecting fake news, their limitations in overfitting, scalability, and generalization impose restrictions on using them in the real world. For practical applications, it is recommended to combine these models with more advanced techniques such as Deep Learning or use Cross-Validation and testing on new data.

7. Ethical and Privacy Considerations in Fake News Detection Systems

However, utilizing an Artificial Intelligence (AI) system for detecting fake news seems useful, but it faces many ethical privacy challenges:

The issue of misinformation labeling and censorship: Fake news detection systems might detect real news as fake news. This error can lead to context censorship, restricting freedom of expression, and avoiding publishing important news. In a sensitive political environment, such a system could be used by governments or corporations to suppress dissenting voices. Algorithmic Bias: AI models are usually trained based on historical data. If that trained data has a racial, political, gender, or cultural bias, the model will learn and reinforce that bias. For example, the probability in this situation will identify news about the minority groups or a specific country as fake. In this project, because the real news comes mostly from Western sources like Reuters, the model might be subconsciously biased toward a particular writing style. Privacy Violations: To train advance model, it is often necessary to collect vast amounts of text data from social media users, news feeds, and personal posts. This can violate users' privacy without their explicit consent. Even if data is anonymized, there are some de-anonymization techniques that can reveal this. Freedom of expression and responsibility: the major risk is that the platforms rely on automated models, which can restrict or eliminate users' context without human intervention. This can show free expression where people refrain from expressing their opinions for fear of being labeled “fake”. Abuse of automated moderation systems: Authoritarian governments or great companies might use this technology to control public information, suppress protests, or shape public opinion. Also, existing adversarial attack like adding small noise to text, can trick the model and make the fake news appear as real news. Overall, the lack of transparency in how the models make decisions prevents users from understanding the reasons why their content is blocked, which reduces public trust (Kaliyar et al., 202).

In terms of harm resulting from privacy violations, individuals and organizations have to consider some legal methods. Data Protection Principles: General Data Protection Regulation (GDPR) in European Union: Individuals have the right to request the deletion of their data (Right to be Forgotten), access to data (Right of Access) and explanation of automated decisions (Right to Explanation), so if the system process the private data without consent intensive financial penalties will apply (European Union, 2016). Personal Data Protection Law in Iran, Malaysia, and other countries: There are similar principles, such as necessity, proportionality, and transparency in data collection, the violation of which is subject to legal action. Legal liability of platforms: affected people can complain to the legal court and claim for their damage. If the misclassification causes financial loss, damage to reputation, or loss of opportunity, compensation can be claimed. In cases of privacy violations, the principles of the right to privacy enshrined in the Universal Declaration of Human Rights (Article 12) and national laws can be invoked. AI Governance Principles: UNESCO and OECD Principles of AI Ethics: here their emphasizes on justice, transparency, fairness, non-discrimination, and responsibility. Also, organizations should conduct an ethical impact assessment (High-Level Expert Group on Artificial Intelligence, 2019; UNESCO, 2021; OECD, 2024). Legislation in the pipeline, such as the EU AI Act: This law requires high-risk AI systems (such as content recognition) to be transparent, human-monitored, and reportable. Violations of these regulations can lead to heavy fines. Practical solutions for individuals: Request an explanation from the platform (Right to Explanation), Complain to data protection authorities (such as the Data Protection Commission), Bring legal action for compensation, Public pressure through the media and human rights organizations (European Union, 2024).

Conclusion: Although fake news detection technology can help combat misinformation, without strong legal frameworks and ethical principles, there is a risk of abuse and violation of individuals' fundamental rights. Therefore, the development of such systems must be accompanied by adherence to the principles of transparency, fairness, accountability, and privacy protection. The governance of AI requires a balance between technical accuracy and human values.

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